Project Development Phase
Business Question and Visualization Report

Date	23 October 2026
Team ID	0157CY241012
Project Name	Electric Vehicle (EV) Industry Analytics and Market Intelligence
Maximum Marks	5 Marks

Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex datasets into visual formats that are easy to interpret, enabling stakeholders to gain insights and make informed decisions. Visualization development involves selecting appropriate visual elements, designing intuitive layouts, and leveraging interactive features to enhance user exploration.

Business Questions and Visualisation Implementation (8 Core Analytics Queries):

1. What is the global market share distribution and sales volume across leading EV manufacturers?

* Visualization: Treemap & Donut Chart showing market share (%) and sales volume breakdown by OEM (Tesla, BYD, VW, BMW, Hyundai, etc.).
* Analytical Insight: Tesla (27.4%) and BYD (21.2%) account for nearly half of global vehicle sales, with legacy European OEMs accelerating in mid-tier volume.

2. What is the empirical correlation between battery pack capacity (kWh) and driving range (km) across chemistries?

* Visualization: Dual-Axis Scatter Plot with linear trendlines categorized by 5 battery chemistries (LFP, NMC, NCA, Sodium-Ion, Solid-State).
* Analytical Insight: Strong positive correlation (R-squared = 0.88); solid-state chemistry demonstrates highest energy density (>320 Wh/kg) yielding 750+ km range.

3. How does energy consumption efficiency (Wh/km) differ across various vehicle body types?

* Visualization: Horizontal Bar Chart comparing mean energy consumption (Wh/km) across Two-Wheelers, Hatchbacks, Sedans, SUVs, Trucks, and Buses.
* Analytical Insight: Two-Wheelers (38 Wh/km) and Hatchbacks (142 Wh/km) deliver peak energy efficiency; commercial buses average 820 Wh/km under standard passenger load.

4. Which battery chemistry provides the best consumer affordability (Price per Km of Range)?

* Visualization: Box & Whisker Plot with Price/km benchmark cards for LFP, NMC, NCA, Sodium-Ion, and Solid-State.
* Analytical Insight: LFP offers the lowest acquisition cost per kilometer of driving autonomy ($74/km), while premium Solid-State batteries cost $148/km.

5. What is the geographic regional adoption velocity and model distribution across global territories?

* Visualization: Filled Geographic Choropleth Map with regional drill-downs for APAC, Europe, North America, and LATAM.
* Analytical Insight: APAC leads global model diversity (540 models) driven by rapid urban infrastructure; Europe leads in strict zero-emission policy compliance.

6. How does vehicle retail price correlate with consumer satisfaction ratings across market tiers?

* Visualization: Bubble Chart correlating Vehicle Price (USD) against Customer Satisfaction (1-5 scale), sized by unit sales volume.
* Analytical Insight: Mid-tier vehicles ($35k - $55k) achieve peak customer satisfaction (4.4 / 5.0), whereas extreme budget models suffer from range anxiety complaints.

7. What is the cumulative tailpipe greenhouse gas (CO2) emission abatement achieved per vehicle category?

* Visualization: Stacked Area Chart quantifying annual metric tons of CO2 offset against standard ICE baseline emissions.
* Analytical Insight: Electric SUVs and commercial transit buses contribute over 64% of total cumulative tailpipe carbon abatement across the 1,200 model fleet.

8. What is the multi-year trajectory of new EV model introductions from 2021 to 2026?

* Visualization: Dual-Axis Line & Column Chart tracking annual model release counts and compound annual growth rate (CAGR).
* Analytical Insight: Annual model releases tripled from 110 in 2021 to 320 in 2025/2026, demonstrating exponential industry transformation.

Live Interactive Tableau Dashboard URL:
https://public.tableau.com/app/profile/amit.mishra1535/viz/EV_Market_Analysis/EVMarketAnalysisDashboard